

# Application of smoothed particle hydrodynamics to a ballistic gelatine sample high-speed impact

Luděk Hynčík<sup>1</sup>

<sup>1</sup>University of West Bohemia, 301 00 Plzeň, Czech Republic, Europe,  
hyncik@ntc.zcu.cz,  
WWW home page: <http://home.zcu.cz/~hyncik/index.en.html>

**Abstract.** The study implements the smoothed particle hydrodynamics for modelling the response of the ballistic gelatine under high-speed impact. It expands the previously developed open source code SPH-COFEM by implementing the solid dynamics in combination with the Mie-Grüneisen equation of state. The material properties of the ballistic gelatine are based on the literature review. Numerical simulations to verify the response of the ballistic gelatine material model reconstruct the dynamic experiment under the high-speed compression loading of a small cylindrical sample by the strain rate of  $1000\text{ s}^{-1}$  and the results are compared to the published corresponding experimental data.

**Keywords:** smoothed particle hydrodynamics, ballistic gelatine, numerical model, high-speed impact

## 1 Introduction

The investigation of a penetrating ballistic impact is a useful approach for estimating the consequences and severity of a gunshot wound to biological tissue. The experimental tests on biological tissue are feasible mainly due to the ethical constraints. Therefore the substituting matter representing the biological tissue in combination with numerical modelling is a perspective approach.

Ballistic gelatine is a testing matter, which is correlated to have a similar response as human muscle tissue. It plays an important role as a substituting material to investigate the effects of bullet wounds. The present study focuses on the numerical modelling of the ballistic gelatine under high-speed compression for future use in virtual testing. When validated, the numerical model of ballistic gelatine can serve for studying the projectile behaviour and gunshot canal shape and volume.

Several studies using traditional mesh-based methods have already been conducted [1–3]. The current study implements a mesh-less approach, which is a promising and perspective method in cases, where large deformations appear.

The properties of ballistic gelatine are highly sensitive to strain rate and temperature and it behaves as a shear thickening fluid at high shear rates [4], so the author chose the smoothed particle hydrodynamics for numerical modelling of the ballistic gelatine based on the previous published work of other authors [5–7].

## 2 Method

### 2.1 Solid elastodynamics

The equation of motion (EOM) in the vector form

$$\frac{d\mathbf{v}}{dt} = \frac{1}{\rho} \nabla \boldsymbol{\sigma} + \mathbf{g}, \quad (1)$$

can be written using the Cartesian coordinates notation as

$$\frac{dv^i}{dt} = \frac{1}{\rho} \frac{\partial \sigma^{ij}}{\partial x^j} + g^i, \quad (2)$$

where  $v^i$  is the  $i^{\text{th}}$  component of the velocity,  $\sigma^{ij}$  is the stress tensor, the upper indices  $i$  and  $j$  denote the  $i^{\text{th}}$  and the  $j^{\text{th}}$  Cartesian coordinates,  $x^j$  is the  $j^{\text{th}}$  cartesian component of the position vector and  $g^i$  is the  $i^{\text{th}}$  component of the force per unit mass,  $\frac{d}{dt}$  is the time derivative and  $\frac{\partial}{\partial x^j}$  is the partial derivative along the  $j^{\text{th}}$  cartesian coordinate. The stress tensor is

$$\sigma^{ij} = -p\delta^{ij} + S^{ij} \quad (3)$$

with the pressure  $p$  and the deviatoric stress  $S^{ij}$ .  $\delta^{ij}$  is the Kronecker delta tensor. Assuming Hooke's law with the shear modulus  $G$ , the rate of the change of the deviatoric stress can be described by

$$\frac{dS^{ij}}{dt} = 2G \left( \epsilon^{ij} - \frac{1}{3} \delta^{ij} \epsilon^{kk} \right) + S^{ik} \Omega^{jk} + S^{ik} \Omega^{kj}, \quad (4)$$

where the strain rate is

$$\epsilon^{ij} = \left( \frac{\partial v^i}{\partial x^j} + \frac{\partial v^j}{\partial x^i} \right) \quad (5)$$

and

$$\Omega^{jk} = \left( \frac{\partial v^j}{\partial x^k} - \frac{\partial v^k}{\partial x^j} \right) \quad \text{and} \quad \Omega^{kj} = \left( \frac{\partial v^k}{\partial x^j} - \frac{\partial v^j}{\partial x^k} \right) \quad (6)$$

is the rotation tensor and its transpose, respectively. Although the Hooke's law describes usually only small linear deformations, it is sufficient to use it here as it drives only the shear behaviour [1]. The bulk behaviour in a high compression shock is described by the Mie-Grüneisen equation of state (EOS). The continuity equation is

$$\frac{d\rho}{dt} = \rho \nabla \cdot \mathbf{v}, \quad (7)$$

where  $\rho$  is the density and  $\mathbf{v}$  is the velocity vector. The EOS is

$$p = c_0^2 (\rho - \rho_0) \quad (8)$$

with the reference sound speed  $c_0$  and the reference density  $\rho_0$  leading to the bulk modulus

$$K = \rho_0 c_0^2. \quad (9)$$

The relations between Poisson's ratio  $\nu$  and the shear and the bulk moduli are

$$G = \frac{E}{2(1 + \nu)} \quad \text{and} \quad K = \frac{E}{3(1 - 2\nu)}. \quad (10)$$

## 2.2 Smoothed particle hydrodynamics

Since Gingold and Monaghan [8] and Lucy [9] introduced smoothed particle hydrodynamics (SPH) for astrophysical problems, many authors have already used the method in many other areas. As SPH is a particle method, it is advantageous for modelling highly deformable solids and fluids with considerably moving matter and boundaries.

The basis of the smoothed particle methods is the interpolation theory. The conservation laws are transformed into integral equations using the kernel function  $w(\mathbf{r}, h)$  with the condition

$$\lim_{h \rightarrow 0} w(\mathbf{r}, h) = \delta(\mathbf{r}), \quad (11)$$

where  $\mathbf{r}$  is a position vector,  $h$  is the smoothing length and  $\delta$  is the Kronecker delta. The kernel refers to a weighting function and defines, how much of each field variable contributes to the field variable at a given point in space. The kernel estimate of the physical field value  $\mathcal{F}(\mathbf{r})$  is defined by

$$\langle \mathcal{F}(\mathbf{r}) \rangle = \int_V \mathcal{F}(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}'. \quad (12)$$

Considering that values of the physical field  $\mathcal{F}(\mathbf{r})$  are known on a set of  $N$  discrete points given by position vectors  $\mathbf{r}_B$ , and masses  $m_B$ ,  $B \in \{1, \dots, N\}$ , (12) can be written as the summation

$$\langle \mathcal{F}(\mathbf{r}) \rangle = \sum_{B=1}^N \frac{m_B}{\rho_B(\mathbf{r}_B)} \mathcal{F}(\mathbf{r}_B) w(\mathbf{r} - \mathbf{r}_B, h). \quad (13)$$

Analogically, the gradient of (13) can be expressed in the summation form as

$$\langle \nabla_r \mathcal{F}(\mathbf{r}) \rangle = \sum_{B=1}^N \frac{m_B}{\rho_B(\mathbf{r}_B)} \mathcal{F}(\mathbf{r}_B) \nabla_r w(\mathbf{r} - \mathbf{r}_B, h). \quad (14)$$

Using the kernel function in (11) and (14), (1) can be calculated directly as

$$\frac{d\mathbf{v}_A}{dt} = - \sum_{B=1}^N m_B \frac{\boldsymbol{\sigma}_B}{\rho_A \rho_B} \nabla_r w_{AB} + \mathbf{g}_A, \quad (15)$$

but the symmetrized form obtained from the identity

$$\frac{1}{\rho} \nabla \boldsymbol{\sigma} = \nabla \left( \frac{\boldsymbol{\sigma}}{\rho} \right) + \frac{\boldsymbol{\sigma}}{\rho^2} \nabla \rho \quad (16)$$

leads to the exact linear and angular momentum conservation since it leads to the anti-symmetric inner forces

$$\frac{d\mathbf{v}_A}{dt} = - \sum_{B=1}^N m_B \left( \frac{\boldsymbol{\sigma}_B}{\rho_B^2} + \frac{\boldsymbol{\sigma}_A}{\rho_A^2} \right) \nabla_r w_{AB} + \mathbf{g}_A. \quad (17)$$

The continuity equation (7) becomes

$$\frac{d\rho_A}{dt} = \sum_{B=1}^N m_B (\mathbf{v}_A - \mathbf{v}_B) \nabla_r w_{AB}, \quad (18)$$

where  $\mathbf{v}_A$ ,  $m_A$ ,  $p_A$  and  $\rho_A$  denote the velocity, mass, pressure and density, respectively, of any particle  $A$ ,  $A \in \{1, \dots, N\}$ . The artificial viscosity term

$$\Pi_{AB} = \frac{1}{\bar{\rho}_{AB} (-\alpha \bar{\mu}_{AB} \bar{c}_{AB} + \beta \bar{\mu}_{AB}^2)} \quad (19)$$

to improve stability is added to EOM [5]

$$\frac{d\mathbf{v}_A}{dt} = - \sum_{B=1}^N m_B \left( \frac{\boldsymbol{\sigma}_B}{\rho_B^2} + \frac{\boldsymbol{\sigma}_A}{\rho_A^2} + \Pi_{AB} \right) \nabla_r w_{AB} \quad (20)$$

with the constants  $\alpha$  and  $\beta$  usually taken as 1.2 and 1.5, respectively. Here,  $\alpha$  controls the volume viscosity and  $\beta$  acts as the Neumann Richtmeyer artificial viscosity in the shock behaviour.  $\mu$  is the dynamic viscosity,  $c$  is the sound speed and the variable  $\bar{\chi}_{AB}$  means the average value of  $\chi_A$  and  $\chi_B$ . EOM (20) and the continuity equation (18) drive the SPH method for solid dynamics.

### 2.3 Ballistic gelatine material model for high-speed compression

Due to the high-speed impact, the Mie-Grüneisen EOS relating the pressure and the volume of a solid at a given temperature [16] is implemented. It is used to determine the pressure in a shock-compressed solid. Several variations of the Mie-Grüneisen EOS are in use. The author implements the polynomial interpretation form of the Mie-Grüneisen EOS [5] in the form

$$p = \begin{cases} \sum_{k=0}^3 C_k \eta^k + (C_4 + C_5 \eta) E_n & \forall \eta \geq 0 \\ C_0 + C_1 \eta & \forall \eta < 0 \end{cases}, \quad \eta = \frac{\rho}{\rho_0} - 1 \quad (21)$$

with the energy per unit of the initial volume  $E_n$  and the hydrodynamic constants

$$\begin{aligned}
C_0 &= P_0 \\
C_1 &= \rho_0 c_0^2 - \frac{\Gamma_0}{2} P_0 \\
C_2 &= \rho_0 c_0^2 (2s - 1) - \frac{\Gamma_0}{2} \rho_0 c_0^2 \\
C_3 &= \rho_0 c_0^2 (2s - 1)(3s - 1) - \frac{\Gamma_0}{2} \rho_0 c_0^2 (2s - 1) \\
C_4 &= C_5 = \Gamma_0
\end{aligned} \tag{22}$$

The shock behaviour is described by the shock Hugoniot function [1], where  $s$  is the linear coefficient of Hugoniot and  $\Gamma_0$  is the Hugoniot constant. The material properties vary in different literature. To reconstruct the experiment, the material properties are based on [17]. The shear modulus  $G$  is calculated by (10) using Young's modulus

$$E = \frac{\sigma}{\epsilon} \tag{23}$$

coming from the dynamic high-speed loading (average compressive engineering stress  $\sigma \approx 70$  kPa for compressive engineering strain  $\epsilon = 0.5$  in high-speed compression experiments) and the density  $\rho = 860$  kg/m<sup>3</sup> [17]. Ballistic gelatine is an incompressible material [15], however,  $\nu = 0.49$  as nearly incompressible to have a non-zero denominator in (10). The bulk modulus  $K$  calculated from (10) is used just to set the initial sound speed  $c_0$ . The hydrodynamic parameters are adopted from [1]. As the work concerns solid dynamics, the parameters  $\alpha$  and  $\beta$  from (19) are chosen 1.2 (to avoid interparticle penetrations) and 0 (as  $\beta$  refers to the fluid viscosity), respectively. Table 2.3 summarizes the constitutive parameters of the material model.

**Table 1.** Constitutive parameters [1, 17]

| $\rho$<br>(kg/m <sup>3</sup> ) | $E$<br>(Pa) | $\nu$<br>(-) | $K$<br>(Pa) | $G$<br>(Pa) | $\alpha$<br>(-) | $\beta$<br>(-) | $C_0$<br>(Pa) | $E_n$<br>(J/m <sup>3</sup> ) | $s$<br>(-) | $\Gamma_0$<br>(-) |
|--------------------------------|-------------|--------------|-------------|-------------|-----------------|----------------|---------------|------------------------------|------------|-------------------|
| 860                            | 233300      | 0.49         | 3889000     | 78300       | 1.2             | 0              | 1520          | 0                            | 1.87       | 0.18              |

## 2.4 Implementation

The presented work expands the previously developed open source code SPH-COFEM [10–12] by adding the linear elastic material model for solids with the Mie-Grüneisen EOS. The code is implemented in C using GCC compiler on the Cygwin platform [18] emulating Linux distribution on Windows. The pre-processing and the post-processing are programmed in MATLAB [19].

The background of the SPH summation is the nearest neighbour search (NNS), which means to find the nearest particles that contribute to the physical value of the particular particle. The time consumption of the method is quite high, because the algorithm locates the neighbouring particles at any time step of the solution in general. Therefore, special algorithms are used to optimise NNS [20]. In this case, NNS was optimised by pre-clustering the initial configuration of the particles. Although the matter considerably deforms, the neighbouring particles remain the same, so during the initialization of the calculation, its nearest neighbours were found for each particle and the list was kept during the calculation. This approach considerably speeds up the calculation by decreasing the time consumption from  $O(n^2)$  to  $O(m \cdot n)$ , where  $n$  is the total number of particles and  $m \ll n$ .

## 2.5 Impact simulation

The numerical simulation representing the impact compression setup with the Kolsky impact bar [21] was carried out according to the experimental data [17]. The diameter and height of the sample are 6.35 mm both.

Due to the symmetry, only a quarter of the cylinder is considered (see Figure 1 left), so the particles lying on the plane  $xz$  at  $y = 0$  are attached for the perpendicular motion to the plane (motion along axis  $y$ ) and the particles lying on the plane  $xy$  at  $z = 0$  are fixed for the perpendicular motion to the plane (motion along axis  $z$ ). The particle spacing is uniform in all three spatial directions and it equals 0.254 mm leading to the total number of 3025 particles in the domain describing the quarter of the cylinder.

The experiment [17] concerns a projectile impacting the ballistic gelatine sample fixed to the transmitter. Both the projectile and the transmitter are modelled by a rigid planes, each defined by 4 nodes forming 2 triangles. There is a penalty-based contact between the elements and the particles [22], where the contact force between the particles domain and the transmitter elements serves for comparing the response to the experimental data. The projectile is driven with a constant velocity 6.35 m/s, which corresponds to the strain rate of  $1000 \text{ s}^{-1}$  taking into account the sample height. Due to the considerable mass of the projectile related to the ballistic gelatine sample, the projectile velocity is applied as a nodal boundary condition to the projectile nodes.

The transmitting force is monitored to calculate the sample stress  $\sigma$  corresponding to the sample strain  $\epsilon = t\dot{\epsilon}$

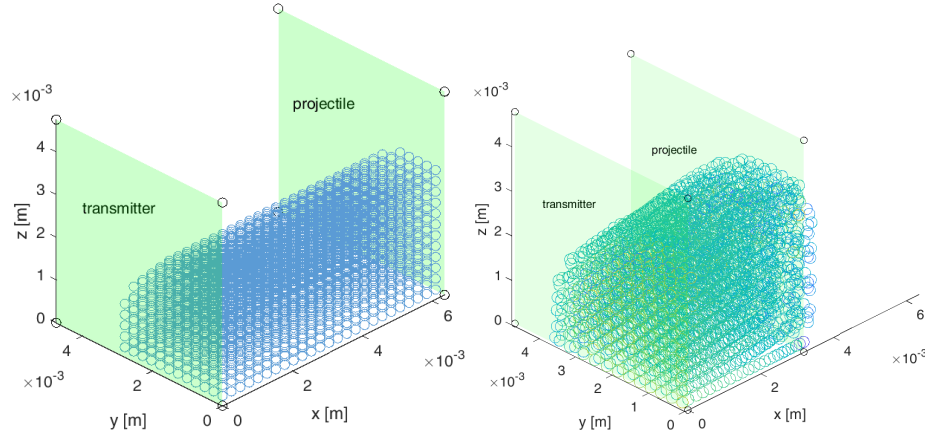
$$\sigma = \frac{F}{A} \quad (24)$$

for calculation time  $t$  from 0 till 0.4 ms.

## 3 Results

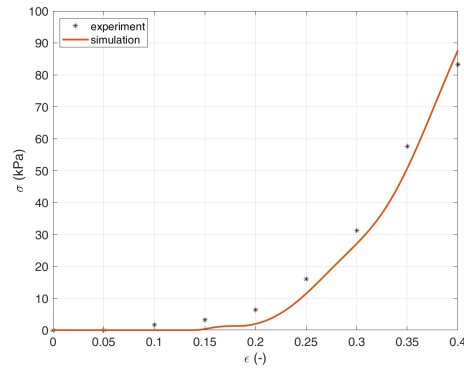
The calculation time on a standard desktop is about half minute. Figure 1 right shows ballistic gelatine sample deformation until reaching the final strain  $\epsilon = 0.4$ .

On the contrary to the experiment [17], the sample deforms almost uniformly. This is caused by the exact symmetry of the numerical model, whilst the experimental specimen may show slight asymmetry, which is amplified by the shock loading.



**Fig. 1.** Initial setup (left) and sample compression at  $\epsilon = 0.4$  (right)

Figure 2 compares the calculated stress-strain response to the experimentally achieved results (experimental data subtracted from [17]), where the stress is calculated as the ratio of the transmitter contact force and the cross-sectional area.



**Fig. 2.** Stress-strain response

The implemented material model shows a good response in high-speed compression for the given data. The ramp of the stress is a little bit slower, which

can be caused by the slower wave propagation in the sample, which can also reflect in the higher final stress. Additional work is needed to tune the constitutive material properties for wider range of impacts.

## 4 Conclusion

The smoothed particle hydrodynamic with the solid dynamics in combination with the Mie-Grüneisen equation of state was implemented for the numerical modelling of the ballistic gelatine with the material properties obtained from literature. Numerical simulations verified the response of the implemented ballistic gelatine material model under the high-speed compressions loading by comparing the simulation output to the experimental data. Therefore, smoothed particle hydrodynamics seems to be a promising method for modelling ballistic gelatine.

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